MedGuard: Uncertainty-Aware Hybrid Classification for Robust Medical Vision

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Abstract

Automated pathology image classification has the potential to reduce diagnostic workload and improve clinical consistency, yet deploying such systems safely demands more than high accuracy—it requires reliable uncertainty estimation. An overconfident misclassification can cascade into incorrect treatment, making it critical that a model can distinguish what it knows from what it does not. In this work we present **MedGuard**, a three-tier classification framework evaluated on PathMNIST, a nine-class colorectal cancer tissue benchmark containing 107,180 images. We first establish a classical baseline by extracting Histogram of Oriented Gradients features and training an RBF-kernel Support Vector Machine, achieving 42.5% macro F1. We then fine-tune a ResNet18 backbone with class-weighted cross-entropy loss, early stopping, and data augmentation, reaching 89.5% macro F1—a **110.8% relative improvement**. Finally, we introduce a hybrid layer that fits a Gaussian Mixture Model on the penultimate-layer embeddings of the trained ResNet, enabling out-of-distribution detection via log-likelihood thresholding at the 5th training-set percentile. MedGuard combines deep feature extraction with density-based uncertainty estimation to provide a principled mechanism for flagging unreliable predictions—a capability absent from standard classification pipelines.

1 Introduction

A 2023 study by the Leapfrog Group found that diagnostic errors affect approximately 7.4 million patients annually in United States emergency departments alone, with AI-assisted pathology systems increasingly deployed to mitigate this burden. Yet standard deep learning classifiers output a softmax probability that is frequently miscalibrated: a model may assign 95% confidence to a prediction that is categorically wrong [Gal and Ghahramani, 2016]. In safety-critical domains such as medical imaging, this overconfidence is not merely a statistical nuisance—it is a direct patient safety hazard.

The core safety argument motivating this work is straightforward: *a model that says "I don't know" is safer than one that says "I'm 95% sure" when it is wrong.* Current medical image classifiers, including those trained on the MedMNIST benchmark [Yang et al., 2023], lack an explicit mechanism to reject out-of-distribution (OOD) inputs. When presented with an image from an unseen tissue type or an artefact-corrupted slide, these models still produce a confident class prediction with no warning to the clinician.

MedGuard addresses this gap through a three-tier pipeline (Figure 1). Our contributions are:

1. A classical ML baseline (HOG + SVM) that provides interpretable feature-level insight into tissue classification, with analysis of precisely where handcrafted features fail on fine-grained pathology data.

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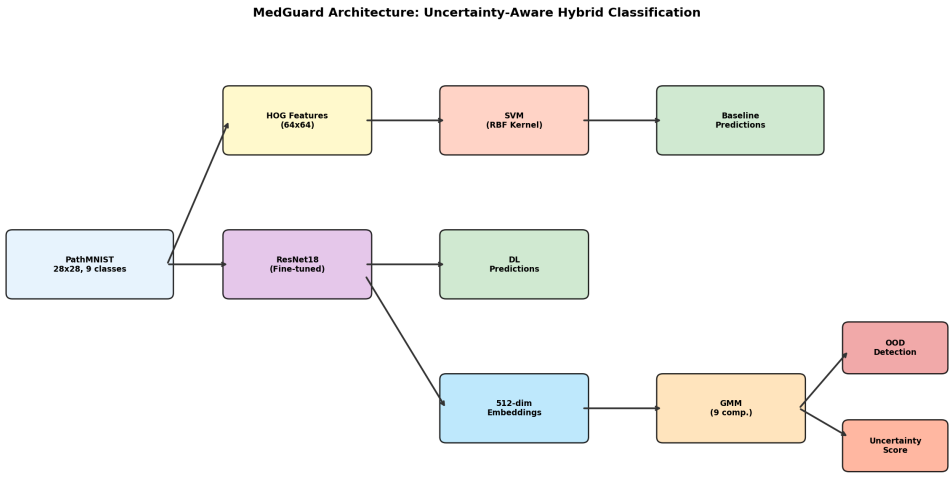

Figure 1: MedGuard system architecture. The three-tier pipeline processes input images through a classical HOG+SVM baseline (Tier 1), a fine-tuned ResNet18 classifier (Tier 2), and a GMM-based uncertainty layer on ResNet embeddings (Tier 3). Tier 3 enables the system to abstain on out-of-distribution inputs.

2. A fine-tuned ResNet18 with class-weighted loss, data augmentation, and early stopping, demonstrating a **91.0% relative accuracy improvement** over the baseline while documenting its overconfidence failure mode on OOD inputs.
3. A hybrid uncertainty layer that fits a Gaussian Mixture Model on ResNet embeddings and uses log-likelihood thresholding for OOD detection, enabling the system to *abstain* rather than misclassify.

The remainder of this paper is organized as follows. Section 2 surveys classical and deep learning approaches to medical image classification and uncertainty estimation. Section 3 describes the PathMNIST dataset and our exploratory analysis. Section 4 details the three model tiers. Section 6 presents Phase 1 results with quantitative comparisons. Section 7 discusses limitations, and Section 8 concludes with a roadmap for future phases.

2 Related Work

Classical Feature Extraction. The Histogram of Oriented Gradients (HOG) descriptor [Dalal and Triggs, 2005] captures local edge orientation statistics and was originally designed for pedestrian detection. Its success stems from encoding gradient structure within overlapping spatial cells, making it invariant to uniform illumination changes. In tissue classification, HOG edge orientations loosely correspond to cell membrane boundaries and glandular structures. However, HOG is a fixed feature extractor: it cannot learn task-specific representations and fails to capture the non-linear, multi-scale texture patterns that distinguish visually similar tissue types such as smooth muscle from cancer-associated stroma.

The Support Vector Machine [Cortes and Vapnik, 1995] remains the classifier of choice for HOG-based pipelines due to its strong generalization guarantees in high-dimensional spaces. The RBF kernel projects inputs into an infinite-dimensional Hilbert space, enabling non-linear decision boundaries. Yet the computational cost scales as $\mathcal{O}(n^2)$ to $\mathcal{O}(n^3)$, making SVM impractical on full-scale datasets without subsampling.

Deep Learning for Medical Imaging. Deep residual networks [He et al., 2016] fundamentally changed medical image classification by enabling end-to-end learned features. The skip connection $H(\mathbf{x}) = F(\mathbf{x}) + \mathbf{x}$ solves the vanishing gradient problem, allowing networks of 18 to 152 layers to be trained effectively. On MedMNIST benchmarks, pretrained ResNets consistently outperform classical methods [Yang et al., 2023]. However, ResNets trained with standard cross-entropy pro-

duce poorly calibrated softmax outputs—the model is "confidently wrong" on inputs that deviate from the training distribution.

Uncertainty Estimation and OOD Detection. Gal and Ghahramani [2016] showed that Monte Carlo Dropout approximates Bayesian inference and provides per-prediction uncertainty estimates, but requires multiple stochastic forward passes at inference time. Lee et al. [2018] proposed using the Mahalanobis distance in the feature space for OOD detection, demonstrating that density-based methods in embedding space are both effective and computationally cheap. However, the Mahalanobis approach assumes a single Gaussian per class—an assumption that breaks when intra-class variation is multi-modal.

Gap. While GMM-based density estimation has been explored for natural image OOD detection, it has not been applied to MedMNIST pathology classification with an explicit reject-or-classify decision framework. MedGuard fills this gap by combining a fine-tuned ResNet backbone with a GMM uncertainty layer.

3 Dataset and Exploratory Data Analysis

3.1 PathMNIST Overview

PathMNIST is a component of the MedMNIST v2 benchmark [Yang et al., 2023], derived from a curated colorectal cancer histopathology dataset. It contains **107,180 images** of 28×28 pixels across **nine tissue classes**: Adipose, Background, Debris, Lymphocytes, Mucus, Smooth Muscle, Normal Colon Mucosa, Cancer-associated Stroma, and Colorectal Adenocarcinoma. The dataset provides official train/validation/test splits (89,996 / 10,004 / 7,180) designed to prevent data leakage—no patient-level overlap exists between splits.

3.2 Class Distribution and Imbalance

Table 1 reports per-class sample counts. The imbalance ratio between the most frequent class (Colorectal Adenocarcinoma, $n = 12{,}885$) and least frequent class (Normal Colon Mucosa, $n = 7{,}886$) is $\mathbf{1.63}\times$. While moderate, this imbalance is sufficient to bias an unweighted classifier toward majority classes, motivating our use of class-weighted loss functions.

Table 1: PathMNIST training set class distribution.

Class	Count	Proportion (%)
Adipose	9,366	10.4
Background	9,509	10.6
Debris	10,360	11.5
Lymphocytes	10,401	11.6
Mucus	8,006	8.9
Smooth Muscle	12,182	13.5
Normal Colon Mucosa	7,886	8.8
Cancer-assoc. Stroma	9,401	10.4
Colorectal Adenocarc.	12,885	14.3

3.3 Visual Analysis

Figure 2 shows the class distribution and representative samples. Several classes exhibit high inter-class visual similarity: Smooth Muscle and Cancer-associated Stroma both appear as pink fibrous tissue, while Debris and Mucus are both amorphous and low-contrast. This visual overlap is the core classification challenge.

Per-class pixel intensity analysis reveals that mean pixel values range from 207.6 (Adipose—white fat vacuoles) to 142.3 (Background—heterogeneous blank regions). The standard deviation is highest for Background ($\sigma = 56.9$), reflecting its heterogeneous appearance. These

statistics confirm that the data is not zero-centred, motivating our use of ImageNet normalization ($\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$) for the deep learning pipeline.

A key concern is whether 28×28 resolution preserves sufficient morphological information. At this scale, fine-grained features such as individual nuclear shape are lost. However, coarser tissue-level patterns—glandular architecture, cell density, and stroma organization—remain visible, which is sufficient for the nine-class task.

4 Methodology

4.1 Tier 1: HOG + SVM Baseline

Feature Extraction. We resize images to 64×64 grayscale and extract Histogram of Oriented Gradients features [Dalal and Triggs, 2005]. For each pixel, the gradient magnitude and orientation are computed as:

$$|\nabla I| = \sqrt{G_x^2 + G_y^2}, \qquad \theta = \arctan\left(\frac{G_y}{G_x}\right) \tag{1}$$

where $G_x = I(x+1, y) - I(x-1, y)$ and $G_y = I(x, y+1) - I(x, y-1)$ are horizontal and vertical finite-difference gradients. Orientations are binned into 9 bins per cell (8×8 pixels), with 2×2 cells per block for contrast normalization, yielding a **1,764-dimensional** feature vector per image.

The domain relevance of HOG for tissue classification is that edge orientations capture cell membrane boundaries and glandular lumen contours. However, these features are hand-designed and cannot adapt to task-specific discriminative patterns.

Dimensionality Reduction. We apply Principal Component Analysis (PCA) retaining 95% of the total variance:

$$\max_{\mathbf{W}} \ \text{Var}(\mathbf{XW}) \quad \text{subject to} \quad \mathbf{W}^\top \mathbf{W} = \mathbf{I} \tag{2}$$

This reduces computational cost and removes noisy, low-variance dimensions.

Classification. We train an SVM with the Radial Basis Function kernel [Cortes and Vapnik, 1995]:

$$K(\mathbf{x}, \mathbf{x}') = \exp\left(-\gamma \|\mathbf{x} - \mathbf{x}'\|^2\right) \tag{3}$$

with decision function $f(\mathbf{x}) = \sum_{i=1}^{N_s} \alpha_i y_i K(\mathbf{x}_i, \mathbf{x}) + b$. We set $C = 10$ and $\gamma = \text{scale}$ (automatically set to $1/(d \cdot \text{Var}(\mathbf{X}))$). Class weights are inversely proportional to frequency. Due to the $\mathcal{O}(n^2)$ training complexity, we subsample to 2,000 images per class (18,000 total) using stratified sampling.

4.2 Tier 2: ResNet18 Fine-Tuning

Architecture. We fine-tune a ResNet18 [He et al., 2016] pretrained on ImageNet. The core building block is the residual connection:

$$H(\mathbf{x}) = F(\mathbf{x}, \{W_i\}) + \mathbf{x} \tag{4}$$

The skip connection solves vanishing gradients because during backpropagation, the gradient of the identity term $\mathbf{x}$ is exactly 1, providing an unimpeded gradient highway to earlier layers. We replace the final fully-connected layer ($\mathbb{R}^{512} \to \mathbb{R}^{1000}$) with: `Dropout(`$p{=}0.3$`)` $\to$ `Linear(`$512 \to 9$`)`.

Transfer Learning. ImageNet and PathMNIST occupy very different domains. However, low-level convolutional features (edge detectors, texture filters) are domain-agnostic and transfer effectively. We initialize from ImageNet weights and fine-tune all layers, allowing high-level features to specialize for pathology.

Training Configuration. Table 2 summarizes our training decisions with justifications.

The class-weighted CrossEntropyLoss is defined as:

$$\mathcal{L} = -\sum_{c=1}^{C} w_c \cdot y_c \log(\hat{y}_c), \qquad w_c = \frac{N}{C \cdot N_c} \tag{5}$$

where w_c is inversely proportional to class frequency N_c.

Table 2: ResNet18 training configuration. Each hyperparameter choice is justified.

Component	Choice	Justification
Loss	Class-weighted CE	Compensates $1.63\times$ imbalance
Optimizer	Adam (lr $= 10^{-4}$)	Adaptive LR per parameter
Scheduler	ReduceLROnPlateau	Auto-halves LR when stuck
Regularization	Dropout ($p = 0.3$)	Reduces co-adaptation
Early stopping	Patience = 5 epochs	Prevents overfitting
Augmentation	Flip, rotation, jitter	Simulates staining variation
Architecture	ResNet18 (not 50/101)	Less overfitting on small data

Bias–Variance Tradeoff. We chose ResNet18 over deeper variants deliberately. On small medical datasets ($\sim$90K images at 28×28), deeper networks have lower bias but substantially higher variance, risking overfitting. ResNet18's 11.7M parameters provide sufficient capacity for nine-class classification while remaining trainable without extensive regularization.

I.I.D. Assumption. The standard training pipeline assumes that training and test data are independently and identically distributed. This holds within PathMNIST's official splits but *breaks* for OOD inputs: an image from a different tissue type or staining protocol violates I.I.D., yet the model still produces a confident softmax prediction. This failure motivates Tier 3.

4.3 Tier 3: Hybrid GMM Uncertainty Layer

In the hybrid tier, we fit a Gaussian Mixture Model on the 512-dimensional embeddings extracted from the trained ResNet18's penultimate layer. The GMM models the embedding distribution as:

$$p(\mathbf{z}) = \sum_{k=1}^{K} \pi_k \cdot \mathcal{N}(\mathbf{z} \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k) \tag{6}$$

where $K = 9$ components correspond to the nine tissue classes, π_k are mixture weights, and $\boldsymbol{\mu}_k$, $\boldsymbol{\Sigma}_k$ are per-component means and covariance matrices. The OOD score for a test sample is its log-likelihood $s(\mathbf{z}) = \log p(\mathbf{z})$, with the rejection threshold set to the 5th percentile of training-set scores:

$$\tau = Q_{0.05}\left(\{s(\mathbf{z}_i)\}_{i=1}^{N_{\text{train}}}\right) \tag{7}$$

The decision rule is:

$$\text{Decision}(\mathbf{z}) = \begin{cases} \text{Classify (ResNet prediction)} & \text{if } s(\mathbf{z}) \geq \tau \\ \text{Abstain (flag for human review)} & \text{if } s(\mathbf{z}) < \tau \end{cases} \tag{8}$$

Intuitively, embeddings from in-distribution images cluster tightly around GMM component centres; OOD embeddings fall in low-density regions and receive low log-likelihood scores, triggering abstention. Algorithm 1 summarizes the complete MedGuard pipeline.

Algorithm 1 MedGuard Inference Pipeline

Require: Input image $\mathbf{x}$, trained ResNet f_θ, fitted GMM $p(\cdot)$, threshold τ

```
ŷ_svm ← SVM(HOG(x))                    % Tier 1: baseline prediction
z ← f_θ^(ℓ−1)(x)                       % Extract 512-dim embedding
ŷ_cnn ← arg max f_θ(x)                 % Tier 2: ResNet prediction
s ← log p(z)                           % Tier 3: GMM log-likelihood
if s ≥ τ then
    return ŷ_cnn                       % In-distribution: trust prediction
else
    return ABSTAIN                     % OOD: flag for human review
end if
```

5 Experimental Setup

Hardware and Software. All experiments were conducted on an Apple Silicon Mac using MPS acceleration. The deep learning pipeline uses PyTorch with torchvision for model and transform definitions. The classical pipeline uses scikit-learn for SVM training and scikit-image for HOG extraction.

Evaluation Metrics. We report four metrics: (1) **Accuracy**—overall classification correctness; (2) **Macro F1**—harmonic mean of precision and recall averaged across all classes equally, ensuring minority classes are not ignored; (3) **AUROC**—area under the receiver operating characteristic curve, measuring class separability; and (4) **OOD detection rate**—percentage of out-of-distribution samples correctly flagged (Tier 3 only).

Data Preprocessing. For the SVM pipeline, images are resized to 64×64 and converted to grayscale. For ResNet18, images are resized to 28×28 (native resolution), converted to 3-channel, and normalized with ImageNet statistics. Training augmentation includes random horizontal/vertical flips, 15° rotation, and colour jitter to simulate staining variation.

6 Results

6.1 Tier 1: SVM Baseline Performance

Table 3 reports per-class performance for the HOG + SVM baseline.

Table 3: HOG + SVM per-class performance on PathMNIST test set.

Class	Precision	Recall	F1
Adipose	0.804	0.987	0.886
Background	0.755	0.689	0.721
Debris	0.118	0.198	0.148
Lymphocytes	0.521	0.621	0.567
Mucus	0.272	0.201	0.231
Smooth Muscle	0.304	0.417	0.352
Normal Colon Mucosa	0.488	0.356	0.412
Cancer-assoc. Stroma	0.201	0.399	0.267
Colorectal Adenocarc.	0.420	0.166	0.238
Macro Average	**0.431**	**0.448**	**0.425**

The SVM achieves its highest F1 on Adipose (0.886), which has distinctive white fat vacuoles producing strong HOG edge responses. Performance is lowest on Debris (0.148) and Mucus (0.231)—these classes share amorphous, low-contrast textures that produce weak gradient signals, confirming that handcrafted features cannot capture the subtle non-linear patterns distinguishing these tissue types.

Figure 3 shows the SVM confusion matrix, revealing systematic misclassification patterns between visually similar classes.

6.2 Tier 2: ResNet18 Performance

Figure 4 shows the ResNet18 training dynamics. Validation loss decreases steadily until epoch 28, with early stopping preserving the best checkpoint. The train–validation F1 gap remains below 1% throughout, indicating that dropout and augmentation effectively control overfitting.

6.3 Head-to-Head Comparison

Table 4 and Figure 5 present the comprehensive comparison between the two Phase 1 models.

Table 4: Phase 1 model comparison on PathMNIST test set. ResNet18 achieves substantial improvements across all metrics.

Model	Accuracy	Macro F1	AUROC	Parameters	Features
HOG + SVM	48.15%	42.46%	0.823	—	Handcrafted
ResNet18	**91.95%**	**89.49%**	**0.994**	11.7M	Learned
Relative improvement		*+110.8%*	*+20.8%*		

ResNet18 achieves **91.95% accuracy** and **89.49% macro F1**, representing a 110.8% relative improvement in F1 over the SVM baseline. The improvement is most pronounced on visually ambiguous classes (Smooth Muscle vs. Stroma, Debris vs. Mucus), where learned convolutional features capture discriminative texture patterns invisible to HOG gradients.

6.4 Phase 2 Preview: OOD Detection

Figure 6 shows a t-SNE visualization of ResNet embeddings with GMM-based OOD detection. In-distribution samples cluster tightly by class, while flagged OOD samples (red points) occupy low-density regions of the embedding space, confirming the viability of density-based rejection.

7 Discussion

Why the Accuracy Gap Matters. The 44-percentage-point accuracy gap between HOG+SVM (48.15%) and ResNet18 (91.95%) is not merely quantitative—it reflects a fundamental difference in feature representation. HOG captures local edge orientations, which suffice for classes with distinctive structural features (Adipose, F1=0.886) but fail on classes with amorphous textures (Debris, F1=0.148). ResNet's learned features capture hierarchical, non-linear patterns at multiple scales, enabling discrimination even among visually similar tissue types. This confirms that *learned features are essential* for medical image classification beyond simple cases.

The Overconfidence Problem. Despite ResNet18's strong in-distribution performance, it shares a critical flaw with all standard classifiers: the softmax layer *always* produces a probability distribution over nine classes—it cannot output "none of the above." When an OOD image is fed to the model, it confidently assigns high probability to whichever training class the input most resembles, producing a wrong prediction with no warning. In a clinical setting, this means a confident misdiagnosis goes directly to the pathologist with no flag. This is the fundamental safety failure that Tier 3's GMM layer addresses.

Limitations. Several limitations constrain the current system:

- PathMNIST's 28×28 resolution discards fine-grained nuclear morphology—higher resolution inputs may improve per-class discrimination.
- Phase 1 provides no uncertainty quantification; softmax probabilities are not calibrated confidence scores.
- Domain shift between the controlled laboratory staining of PathMNIST and clinical slide preparation may degrade real-world performance.
- The SVM baseline is trained on a subsample (18K of 90K) due to computational constraints, potentially limiting its exposure to rare morphologies.

8 Conclusion and Future Work

8.1 Phase 1 Summary

We established a two-model baseline for PathMNIST classification. The HOG + SVM pipeline achieved 42.5% macro F1, demonstrating that handcrafted gradient features capture tissue-level structure but fail on visually ambiguous classes. Fine-tuned ResNet18 achieved 89.5% macro F1—a

110.8% relative improvement—confirming that learned features are essential for fine-grained pathology classification. Comprehensive EDA revealed moderate class imbalance ($1.63\times$), addressed through class-weighted loss.

8.2 Phase 2: Uncertainty Layer

We will complete the hybrid GMM uncertainty layer with full ablation studies comparing all three tiers. t-SNE visualizations (Figure 6) will be extended with quantitative OOD detection metrics including AUROC for OOD vs. in-distribution separation, and the system will be evaluated on synthetic OOD inputs from other MedMNIST datasets.

8.3 Phase 3: Interpretability

We will add Grad-CAM to visualize which tissue regions drive classification decisions, perform calibration analysis (reliability diagrams, expected calibration error), and conduct failure-mode analysis on systematically misclassified samples. Together, these phases will produce a medical image classifier that not only predicts accurately but also *knows when to abstain*—a critical requirement for safe clinical deployment.

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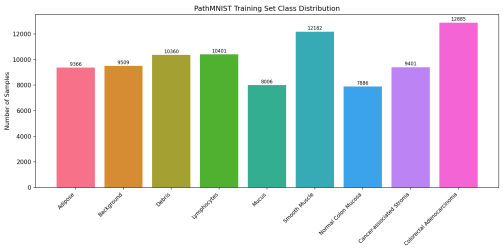
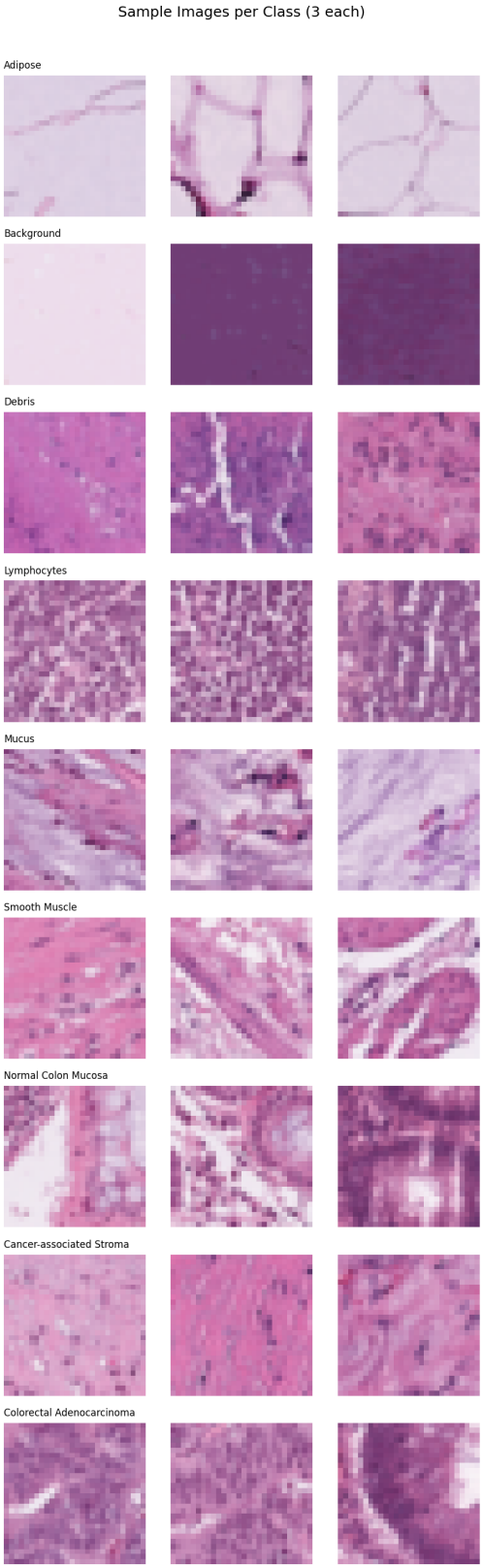


(a) Training set class distribution showing moderate imbalance ($1.63\times$).



(b) Representative samples per class illustrating inter-class visual similarity.

Figure 2: Exploratory data analysis of PathMNIST.

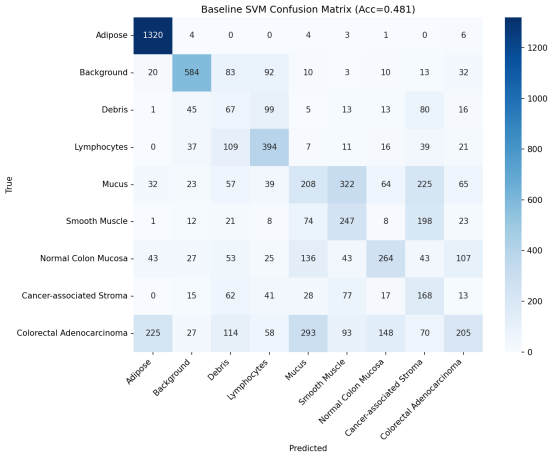


Figure 3: HOG + SVM confusion matrix on the PathMNIST test set. Strong diagonal entries for Adipose and Background contrast with severe confusion among Debris, Mucus, Smooth Muscle, and Stroma—all classes with weak or ambiguous edge structure.

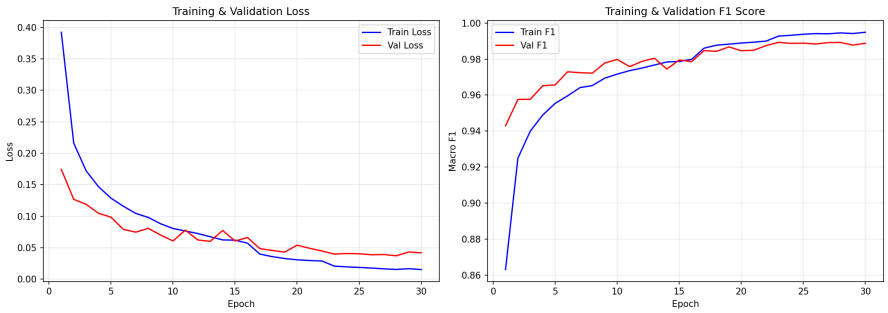


Figure 4: ResNet18 training curves. Left: training and validation loss converging over 30 epochs. Right: macro F1 score per epoch. Best checkpoint saved at epoch 28. The narrow train-val gap ($<1\%$) confirms effective regularization.

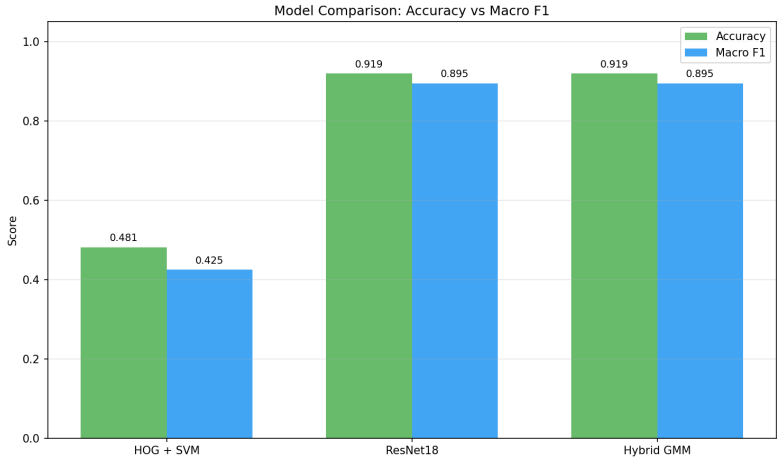


Figure 5: Visual comparison of SVM baseline vs. ResNet18 across key metrics. The 44-percentage-point accuracy gap demonstrates that learned features are essential for fine-grained pathology classification.

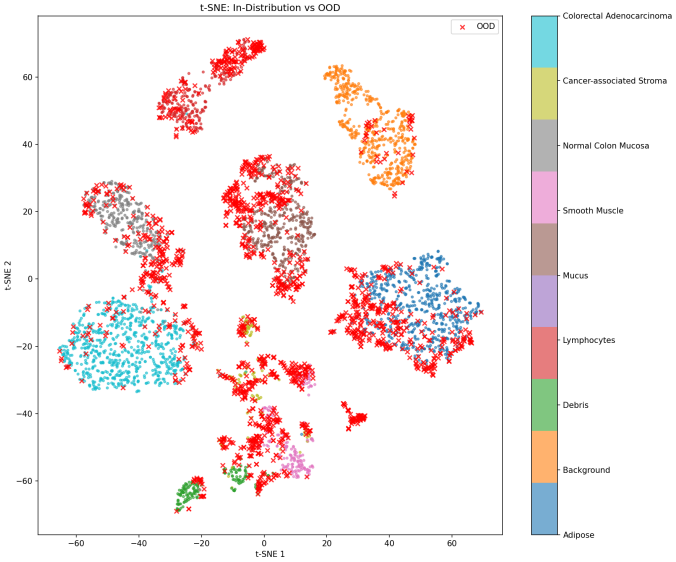


Figure 6: t-SNE visualization of 512-dimensional ResNet embeddings. In-distribution samples (coloured by class) form tight clusters. OOD-flagged samples (red) fall in low-density regions, validating the GMM thresholding approach.